# The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

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## 1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification ($Fx$), while the invariant control forests encode properties atemporally ($xF$). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space ($V'$), culminating in physical vehicle extraction.

## 2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[ \boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

## 3 High-Fidelity Electromagnetic Transient (EMT) Systems and IEEE 39-Bus Testing

To ground the continuous-time physics layer into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus test system via the open-source Julia platform `PowerSimulationsDynamics.jl`. Let the transient state of an electrical node $i$ be characterized by a continuous differential-algebraic power vector $\boldsymbol{\xi}_i(t)$:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

where $V_i(t)$ represents the instantaneous voltage magnitude, $\delta_i(t)$ defines the phase angle deviation, and $P_{\text{IBR},i}(t), Q_{\text{IBR},i}(t)$ monitor the active and reactive power outputs of local inverter-based renewable resources. The sub-millisecond voltage variations are processed natively by the inductive learners as extensional grounding exemplars (`exs.pl`):

$$\mathcal{E}_i \implies \text{Ingest}\left(\boldsymbol{\xi}_i(t_{\text{sample}}),\, \Delta t = 0.5 \text{ ms}\right) \tag{3}$$

This allows our distributed MARL controllers to intercept microscopic transient voltage droops ($V_i(t) < 0.95$ p.u.) and enforce the Sherman-Gable (1983) backoff boundary constraints before sub-synchronous oscillations propagate into catastrophic inter-area blackouts.

## 4 Black-Hunter-Pan Multi-Ontology Dialogue Poset Strategy

When independent corporate data centers must resolve grid access boundaries without data leakage or brute-force database merges, the deconfliction process is modeled as a preference-ordered dialogue game following the algebraic principles of **Black, Hunter & Pan**.

Agents exchange lightweight symbolic arguments formulated as support-claim tuples $\mathcal{A}_k = \langle \Phi, \alpha \rangle$ across a Model Context Protocol (MCP) tool channel. Let $\mathcal{D}_t$ represent the sequential state-history matrix of the active dialogue at turn $t$. The turn-taking strategy function $\mathcal{S}_x(\mathcal{D}_t)$ acts as a deterministic, preference-ordered selection transducer over the Description Logic poset, mapped via the following strict dominance hierarchy:

$$\mathcal{S}_x(\mathcal{D}_t) = \text{Scounter}_x(\mathcal{D}_t) \succ \text{Sarg}_x(\mathcal{D}_t) \succ \text{Sconcede}_x(\mathcal{D}_t) \succ \langle x, \text{close}, \gamma \rangle \tag{4}$$

where:

$$\text{Scounter}_x(\mathcal{D}_t) = \{\langle x, \text{posit}, \langle \Phi, \phi \rangle \rangle \mid \langle \Phi, \phi \rangle \text{ is a novel Undercutter counter-argument}\} \tag{5}$$

$$\text{Sarg}_x(\mathcal{D}_t) = \{\langle x, \text{posit}, \langle \Phi, \gamma \rangle \rangle \mid \gamma = \text{Topic}(\mathcal{D}_t)\} \tag{6}$$

$$\text{Sconcede}_x(\mathcal{D}_t) = \{\langle x, \text{concede}, \phi \rangle \mid \phi \in \Delta_x \cup \Pi \text{ and } \phi \text{ is relevant to } \gamma\} \tag{7}$$

Transmission bandwidth overhead $\mathcal{C}$ scales as a compact function of decision forest graph depth $d$ rather than the raw magnitude of the extensional grounding data:

$$\mathcal{C}(\mathcal{T}) \propto \mathcal{O}(d \cdot \Sigma_{\text{tokens}}) \ll \mathcal{C}(\mathcal{E}) \propto \mathcal{O}(|\mathcal{E}| \cdot \text{Dim}_{\text{features}}) \tag{8}$$

This enables the multi-agent perimeter to achieve absolute strategic consensus $\langle x, \text{close}, \gamma \rangle \cap \langle y, \text{close}, \gamma \rangle$ over thin network channels while keeping proprietary Level 1 telemetry perfectly firewalled.